

GUANGCHANG, China

WMO: 588130

Lat: 26.8379N

Lon: 116.3189E

Elev: 466

StdP: 14.45

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.6	32.7	20.1	15.4	43.6	24.5	18.9	41.9	11.9	46.9	10.5	47.6	1.9	180	0.324

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.5	97.2	79.0	95.4	78.9	93.6	78.5	81.1	91.0	80.5	90.3	79.9	89.2	5.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.0	153.4	84.2	78.3	149.4	83.7	77.7	146.3	83.4	45.4	91.3	44.6	90.5	44.0	89.6	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
11.3	9.6	8.3	DB	26.1	100.3	2.7	1.6	24.2	101.4	22.6	102.3	21.1	103.2	19.2	104.3
			WB	25.3	82.7	2.8	0.9	23.2	83.3	21.6	83.8	20.0	84.3	17.9	84.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	66.9	45.5	50.2	57.1	67.4	75.0	80.3	85.0	84.1	78.9	70.1	59.6	48.8
	DBStd	15.00	7.23	9.12	8.22	7.06	4.84	4.56	3.42	3.43	5.35	5.72	6.98	6.31
	HDD50	416	180	103	26	1	0	0	0	0	0	0	10	96
	HDD65	2060	604	421	270	55	1	0	0	0	0	19	188	502
	CDD50	6597	42	109	246	524	775	910	1085	1056	867	624	299	60
	CDD65	2765	0	8	24	128	311	460	620	591	418	178	26	1
	CDH74	27674	5	66	181	847	2297	4469	7697	6833	3933	1168	172	6
	CDH80	11843	0	6	28	233	772	1803	3841	3173	1631	328	28	0
Wind	WSAvg	3.5	3.4	3.6	3.4	3.4	3.2	3.4	3.7	3.5	3.6	3.6	3.3	3.4
Precipitation	PrecAvg	67.7	2.6	4.4	7.4	9.9	11.5	11.7	4.5	4.3	3.6	3.2	2.7	1.7
	PrecMax	91.9	8.6	10.2	13.1	18.4	24.9	27.3	15.6	15.3	13.0	10.6	9.1	4.9
	PrecMin	44.8	0.3	0.3	2.0	2.8	4.0	4.4	0.7	0.8	0.2	0.0	0.1	0.1
	PrecStd	12.7	2.0	2.2	3.1	3.9	4.8	5.9	3.0	2.8	2.8	2.8	2.2	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.1	80.2	83.4	89.5	93.6	96.1	100.1	99.5	96.9	91.1	83.7	73.1
		MCWB	61.7	65.5	68.4	72.8	74.6	78.8	79.0	79.0	78.3	74.2	68.7	61.5
	2%	DB	66.9	75.6	79.2	85.8	90.5	93.7	97.5	97.1	94.3	87.0	79.1	67.3
		MCWB	57.3	63.7	65.9	71.2	73.7	78.3	79.2	79.1	77.9	72.0	66.7	56.6
	5%	DB	62.0	70.2	74.7	82.6	87.5	91.6	95.7	95.1	91.8	83.7	74.9	63.0
		MCWB	53.8	61.0	64.3	69.9	73.5	78.0	79.2	79.1	77.4	69.5	63.7	54.9
	10%	DB	57.0	64.4	70.3	79.1	84.5	89.3	93.8	92.7	89.0	80.6	71.0	59.5
		MCWB	51.0	57.9	62.8	68.8	73.5	77.7	78.9	79.1	76.3	68.4	62.6	52.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.9	67.8	71.1	75.4	79.2	81.3	82.3	81.9	80.9	76.6	72.6	64.5
		MCDB	69.7	77.4	78.8	84.5	86.6	89.2	92.7	92.4	90.3	85.3	79.5	70.9
	2%	WB	59.2	64.7	68.6	73.6	77.8	80.3	81.2	81.0	79.7	74.6	69.5	60.4
		MCDB	65.3	73.4	75.9	82.0	84.8	88.7	91.6	91.1	89.1	82.0	74.4	64.1
	5%	WB	55.6	61.8	66.3	72.3	76.5	79.6	80.5	80.4	78.8	73.1	67.2	56.7
		MCDB	60.2	69.1	72.7	79.4	82.9	87.6	90.8	90.1	87.9	79.8	71.8	61.3
	10%	WB	51.7	58.6	63.8	70.8	75.4	78.8	79.9	79.8	77.9	71.6	64.3	53.8
		MCDB	56.5	63.9	68.8	76.8	80.9	86.4	89.7	88.9	86.0	77.4	69.7	57.9
Mean Daily Temperature Range	5% DB	MDBR	12.3	13.2	13.6	14.3	14.3	12.8	15.5	15.0	14.9	16.0	15.7	15.0
		MCDBR	23.7	24.7	23.5	21.5	21.0	17.5	19.1	19.4	19.3	21.5	23.6	24.5
		MCWBR	13.7	13.0	10.9	8.2	7.1	4.8	4.5	4.4	4.9	6.9	10.5	13.7
	5% WB	MCDBR	19.2	21.5	18.4	16.9	14.6	14.4	17.0	16.8	17.0	14.9	16.6	17.2
		MCWBR	12.3	11.8	9.7	8.1	6.2	5.1	4.9	4.7	4.8	5.9	8.5	10.7
Clear-Sky Solar Irradiance	taub	0.647	0.694	0.779	0.751	0.651	0.617	0.577	0.625	0.635	0.657	0.605	0.592	
	taud	1.590	1.543	1.441	1.530	1.753	1.839	1.951	1.810	1.762	1.654	1.712	1.699	
	Ebn at Noon	187	192	186	199	220	226	235	223	215	198	197	193	
	Edh at Noon	74	84	98	91	73	67	60	68	70	74	65	64	
All-Sky Solar Radiation	RadAvg	675	780	890	1079	1237	1287	1751	1548	1328	1147	870	784	
	RadStd	157	183	132	162	175	151	145	129	128	196	148	168	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.54	N/A	N/A	N/A	+0.36	+0.58	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	+0.67	N/A	N/A	+0.99	N/A	N/A	N/A	N/A	+252	+178

Nomenclature: See separate page